Winter of Code – Project Listing

No-Code Business Process Management (BPM) Platform: Form, Status, Process and Transition Engine with Approval Engine Integration

Company: Technorishi Consulting Pvt. Ltd.

Duration: 8 weeks

Number of Participants: 2–3 students

Mentor: 1 Technical Project Head / technical mentor

Project Title

No-Code Business Process Management (BPM) Platform: Form, Status, Process and Transition Engine with Approval Engine Integration

Problem Statement

Our platform already has an approval workflow designer and engine that decides who approves what and in which order. What is missing is the layer that defines the business process around that approval. This layer covers three things: the form that captures the data, the stages the request passes through, and the statuses and lifecycle it follows from creation until it closes.

Today, each new process needs development effort to build its form, define its statuses, and wire it to the approval engine. Business users cannot set up or change a process on their own.

We want a no-code layer where an administrator can design forms, define statuses, model processes as stages, and configure the transitions between them. Some transitions will hand control to the existing approval engine and resume based on the approval outcome. Every submitted form should run as a tracked process instance until it reaches a terminal status such as Completed, Rejected, or Cancelled, with a full audit trail.

The challenge has two parts:

1. Design a metadata-driven engine where forms, statuses, processes, and transitions are configuration, not code.
2. Integrate it cleanly with an existing approval engine that the students will not modify.

Project Description

Technorishi builds enterprise platforms across HCM, facility management, compliance, and permit-to-work, all of which need configurable processes. The approval engine is already in place. This project builds the reusable process layer on top of it. The selected students will build four designers and one engine.

Form Designer and Engine

- Configurable form builder with common field types (text, number, date, dropdown, checkbox, file upload).
- Field validations and required/optional rules.
- Form versioning so that edits do not affect in-flight submissions.

- Runtime rendering of forms from their metadata.

Status Designer

- Status master for each process.
- Status names and categories (open, in-progress, terminal).
- Display attributes such as colour and label.

Process Designer

- Define a process as a set of stages.
- Link a form to the process.
- Set the lifecycle from the start state to the terminal states.

Transition Designer and Engine

Defines the allowed transitions between stages and statuses, including:

- Which actions trigger them (Submit, Approve, Reject, Send Back, Cancel).
- Who can perform them (by role or user).
- Simple conditions based on form field values.
- Which events fire on each transition, such as a notification or webhook.

The engine enforces only valid transitions and records the full history.

Approval Engine Integration

A transition can be marked as an approval step. When the process reaches it, the engine will:

1. Invoke the existing approval engine with the request and form data.
2. Hold the process instance in an "awaiting approval" state.
3. Receive the approval outcome through a callback or event.
4. Move to the configured next stage and status based on that outcome.

Students should propose the integration contract. This includes the request payload, how outcomes are returned, how failures are handled, and how process status and approval status stay in sync.

Expected Outcome

A working proof of concept that shows two sample processes with different stages and lifecycles, with at least one of them routed through the existing approval engine. Each process should run end to end:

Design form → Define statuses → Model process → Configure transitions → Submit → Approval handoff and return → Form closes as Completed

Required Skills

- Backend development with REST APIs.
- Relational database design.
- Frontend development.
- Understanding of state machines and system integration (APIs, callbacks or events).

Duration

4 weeks.

Number of Participants

3–4 students. A suggested split of responsibilities:

- Form designer and runtime.
- Status and process designer.
- Transition engine and approval integration.

Mentor

One company-appointed Technical Project Head or technical mentor from Technorishi Consulting Pvt. Ltd.

Eligibility / Prerequisites

- Pre-final or final year students in CSE, IT, or related branches.
- Prior full-stack project experience.

Technology / Tools

Spring Boot (Java) or .NET Core for the backend, React for the frontend, and PostgreSQL or MS SQL for the database. Integration with the approval engine will be through its existing APIs. Students may propose alternative technologies with a justification.

Deliverables

1. Architecture and data-model document, including the approval engine integration contract.
2. Working form, status, process, and transition designers.
3. Process execution engine with transition validation and an audit trail.
4. Working integration with the approval engine (or a provided sandbox of it).
5. Two configured demo processes.
6. Source code repository.
7. API documentation.
8. Demo video and setup guide.

Out of Scope

The following are outside the scope of this four-week proof of concept and may be listed as future enhancements:

- Changes to the existing approval engine or approval designer.
- Parallel stages.
- SLA timers and escalations.
- Multi-tenancy.
- Production-grade security hardening.